OLIVER ERNSTER

Technical
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Decision Architecture (Published Work)

Author of the Decision Architecture series, a four-part published work examining how decision systems, authority structures and organisational design determine engineering outcomes.

Focus areas:

- Decision latency and system behaviour
- Authority boundaries and ownership clarity
- Structural causes of delivery failure

Available via Amazon (Kindle, paperback, hardback)

crankthecode.com

Senior Python Engineer with extensive experience delivering backend platforms and cloud-based systems across defence, fintech, telecoms and startup environments.

I specialise in building and modernising production-grade Python services using FastAPI, Flask and Django, along with Javascript frameworks, with a strong focus on reliability, maintainability and operational clarity.

I have led technical delivery across distributed teams and also worked hands-on as a senior individual contributor, taking ownership of backend architecture, API design, cloud infrastructure and deployment pipelines.

I actively leverage agentic AI development workflows (e.g. Roo, Cline; integrated with MCP) and multiple LLM providers including Claude, OpenAI (GPT variants), and Google Flash across both professional and personal work.

I use these tools to accelerate development, improve code quality, explore architectural options and reduce delivery latency while retaining full ownership of system design and production outcomes.

My experience includes:

- Designing and implementing RESTful Python services in regulated and high-availability environments
- Migrating legacy systems to modern Python 3 platforms
- AWS and serverless architectures including Lambda-based systems
- Infrastructure as code using Terraform
- Performance optimisation, scalability planning and production hardening
- Collaborating closely with product managers, stakeholders and cross-functional teams to deliver stable, maintainable systems
- Applied agentic AI tooling (Roo, Cline) to accelerate development workflows and reduce delivery latency
- Practical experience integrating and working with multiple LLM providers (Claude, OpenAI GPT variants, Google Flash)

Career history (most recent first)

Selected AI-assisted personal projects:

- **NarrateX – LLM-assisted text processing and narration platform (https://www.narratex.co.uk)**
- **Stellody – AI-assisted music classification and playlist generation system (in progress)**
- **Extensive use of agentic AI tools across personal projects, predating professional adoption, with portfolio examples at crankthecode.com**

Cisco – Senior Python Developer (Consultancy) – June 2025 through November 2025

- Leveraged agentic AI development tooling (Roo/Cline) with multiple LLMs (Claude, OpenAI GPT, Google Flash) to accelerate system modernisation, refactoring and architectural iteration while maintaining production-grade quality and control
- Used LLMs as development accelerators rather than decision-makers, retaining full responsibility for system design, correctness and production behaviour
- Owned architectural decisions affecting deployment, maintainability and long-term platform stability, including platform upgrades, dependency rationalisation and the introduction of idempotent, automated deployment practices.
- Established a modern React and TypeScript user interface, materially improving usability, stakeholder confidence and the operability of the platform.

Harrier – Lead Python Developer (Startup) – November 2024 through February 2025

- Applied agentic AI workflows (Roo/Cline) with multiple LLM providers (Claude, OpenAI GPT variants, Google Flash) to accelerate backend development, infrastructure definition and iteration speed while retaining full technical ownership and system accountability
- Designed and implemented a serverless Python backend using AWS Lambda and Zappa, supporting data ingestion, processing and reporting pipelines.
- Defined and built the complete AWS infrastructure using Terraform from first principles, enabling repeatable, secure and scalable operation.
- Held full accountability for technical decisions, system behaviour and reliability under real-world operational constraints.

LTI Mindtree – Lead Python Developer (Consultancy) – February 2023 through August 2023

- Led the replacement of a MuleSoft-based API with a modern, maintainable FastAPI Python 3 platform.
- Set architectural direction, coding standards and documentation practices to improve long-term maintainability and team effectiveness.
- Produced a simplified and lower-cost API solution, reducing operational overhead while preserving continuity for downstream consumers.

EPAM – Lead Developer (Consultancy) – August 2022 through November 2022

- Acted as technical lead for the design and prototyping of a cloud-based flood insurance API.
- Influenced architectural direction and delivery approach within a regulated, multi-stakeholder environment.

Thales – Senior Developer (Cyber Resilience & Defence Research) – September 2021 through December 2021

- Engineered a multithreaded, socket-based backend with a Flask front end to support cyber resilience modelling and visualisation in a defence research context.

Unilink – Senior Developer – March 2021 through September 2021

- Contributed to the design and implementation of secure, reliable communication software intended to support family contact and reduce reoffending rates.
- Balanced technical decisions against operational, ethical and regulatory constraints.

Validus Risk Management – Senior Developer (FinTech) – October 2020 through March 2021

- Contributed to a regulated wealth management and risk analysis platform, delivering enhancements aligned with analytical requirements and business objectives.

Memset – Senior Developer (Cloud Platform) – August 2019 through October 2020

- Enhanced core cloud platform capabilities and participated in platform-level architectural discussions within a pure cloud engineering team.

Hewlett Packard – Senior Developer / Project Technical Lead / Scrum Master – December 2012 through February 2014

- Led a 17-person, multi-disciplinary engineering team delivering a large-scale defence programme.
- Held responsibility for technical direction, delivery planning and cross-team coordination.
- Acted as Scrum Master, improving delivery predictability, team coordination and overall programme execution.

Earlier career (selected)

- Held senior engineering roles across defence, fintech, telecoms and startup environments, including Northrop Grumman, Armour Communications, Bink, Duologi and Anite, progressing through technical and leadership roles in secure and regulated contexts.